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Innovative Sensor-based Approach for Predicting Foot Ulcers in Diabetic Patients using Machine Learning Models

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***Abstract*— Foot ulcers, particularly in diabetic patients, are a significant health concern that can lead to severe complications if left untreated. Early detection of foot ulcers is critical for preventing infections, reducing hospitalizations, and improving the quality of life for at-risk individuals. This project aims to develop an automatic detection system for foot ulcers using an integrated approach that combines hardware sensors and predictive analytics software. The hardware consists of a foot temperature sensor and a pressure sensor to monitor the conditions that contribute to ulcer formation. The software, powered by machine learning algorithms, collects, preprocesses, and analyzes the data from these sensors to build a predictive model capable of detecting early signs of foot ulcers. By leveraging this integrated system, the solution provides a non-invasive, real-time detection method that can be deployed in clinical settings or as part of home monitoring systems for high-risk patients. This technology has the potential to enhance early diagnosis, reduce complications, and lower healthcare costs by enabling timely interventions. In this various predictive model like support vector machine, naïve bayes, decision tree, and random forest has been implemented successfully and the results are compared. This technology has the potential to enhance early diagnosis, reduce complications, and lower healthcare costs by enabling timely interventions.**

Keywords—real-time data analysis, healthcare technology, alert mechanism, Foot ulcer detection, predictive models, SVM, random forest, decision tree, naïve bayes

I. Introduction

Foot ulcers are a significant and common complication, particularly in individuals with diabetes, and can lead to severe consequences if not detected and treated promptly. A foot ulcer is an open sore or wound that typically occurs on the bottom of the foot, often due to poor circulation, nerve damage (neuropathy), and pressure or injury. These ulcers can vary in severity, from minor skin breakdown to deep wounds that expose tendons or bones, and they can quickly become infected. In extreme cases, untreated ulcers may necessitate amputations, particularly for diabetic patients, due to the challenges of healing and controlling infections.

One of the main risk factors for foot ulcer development is diabetes, one of the biggest worldwide health issues. Diabetes frequently causes neuropathy, which impairs foot feeling. As a result, tiny wounds or pressure points may go undetected until an ulcer has developed. Diabetes can also lead to poor circulation, especially in the extremities, which slows the healing process and increases the risk of infection. Diabetic foot ulcers are a major source of morbidity and a leading cause of non-traumatic lower limb amputations worldwide, affecting 15% of individuals with diabetes at some time in their life, according to statistics.

Early detection and intervention are crucial in managing and preventing the progression of foot ulcers. However, conventional methods for detecting foot ulcers, such as physical examinations or relying on patients' self-awareness of symptoms, can be insufficient due to delayed symptom manifestation, especially in individuals with diminished sensation. As such, there is a pressing need for more advanced, automated, and non-invasive methods for early detection and monitoring.

In response to this need, modern technological advancements, including predictive analytics, machine learning, and integrated hardware solutions, have emerged as promising tools for improving the detection and management of foot ulcers. By leveraging data from sensors that monitor foot temperature and pressure, along with predictive models that can analyze trends and predict ulcer formation, healthcare providers can intervene earlier and prevent complications. The integration of hardware and software in this context offers a real-time, proactive approach to monitoring foot health, potentially improving patient outcomes and reducing the healthcare burden associated with diabetic foot ulcers.

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The following sections explore the design and implementation of an automatic foot ulcer detection system that combines hardware sensors with a predictive analytics model, showcasing the advantages and potential impact of this integrated method.

Figure 1:Sample foot Foot ulcer scrren-1

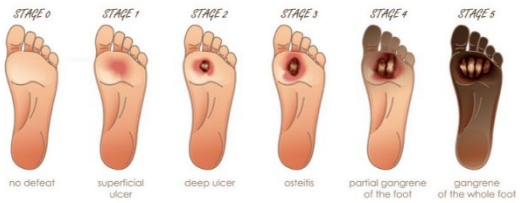

Figure 2:Sample foot ulcer screen-2

II. Literature Review

Foot ulcers are a major healthcare concern because of their high occurrence and potential for serious consequences including infection and amputation, especially in diabetic patients. Researchers have worked to provide better ways to identify, treat, and prevent foot ulcers throughout the last few decades. Because diabetes patients often have neuropathy, or lack of feeling, routine physical checks are the mainstay of traditional detection procedures, and they can occasionally miss early-stage ulcers. This has increased interest in automating and enhancing the identification of foot ulcers by utilizing technology breakthroughs, including predictive analytics, machine learning, and sensor-based systems. The literature analysis that follows highlights significant advancements in this field, with a focus on predictive analytics models that are combined with hardware solutions.

In [1] the context of diabetic foot ulcers (DFU), predictive analytics can provide a proactive solution by analyzing patterns of foot temperature and pressure distribution to predict the onset of ulcers before they become visible. Studies have demonstrated that anomalies in foot temperature and pressure points can serve as early indicators of ulcer formation (Kavitha et al., 2014).

Golledge et al. [2] meta-analysis discovered that at-home foot temperature monitoring significantly lowers the ri sk of diabetic foot ulcers (DFUs). Participants utilizing this method had a 49% lower risk compared to controls. Sensitivi ty analyses validated these findings, despite the overall certai nty being rated low. This highlights the effectiveness of foot temperature monitoring as a preventive measure for DFUs.

Lavery et al.[3]assessed temperature monitoring's role in reducing the recurrence of diabetic foot ulcers (DFUs) in hig h-risk patients. The study concluded that consistent self-monitoring of foot temperature, paired with suitable intervent ions, significantly lowered the rate of secondary ulcer recurre nce.

This study [4] aimed to conduct a comprehensive and current systematic review and meta-analysis of randomized controlled trials (RCTs) to evaluate the effectiveness of home foot temperature monitoring, patient education, and offloading footwear in reducing the incidence of diabetes-related foot ulcers.

Diabetes-related foot disease (DFD) [5] is a major contributor to disability worldwide. It includes ulcers, infections, and gangrene. Since recurrence within a year is typical, long-term care is essential. In order to avoid DFD, this review examines wearables, telehealth, and sensors. Technologies that have the potential to decrease in-person consultations include telemedicine and foot temperature and glucose monitors. Even though studies show that some remote approaches are beneficial, more work has to be done, particularly in the area of plantar pressure monitoring. There is a lack of a complete program for remote management, which emphasizes the necessity for research into holistic remote techniques to avoid DFD.

The study by Ming et al. [6]outlines a randomized control led trial protocol to evaluate the preventive effects of telemed icine on diabetic foot ulceration1. The trial tests sensor-equipped insoles combined with photo documentation to dete ct early signs of ulceration, aiming to avoid severe complicat ions like amputation or infection. This approach seeks to offe r a less intrusive and more costeffective alternative to freque nt clinical screenings

Drovandi et al. [7] provide a systematic review of the effectiveness, stakeholder perceptions, and cost-efficiency of remotely delivered healthcare for diabetes-related foot disease (DFD). The review emphasizes that remote monitoring and management help reduce care disparities, particularly for individuals in remote or underserved areas. By synthesizing evidence from multiple systematic reviews, it highlights the potential of telehealth and sensor-based interventions to effectively manage DFD, though additional research is recommended to refine and enhance these methods.

Minty et al. [8] explore the implementation of sensor-based remote monitoring (RPM) to prevent diabetic foot ulcers (DFUs) in clinical practice. Their study outlines a practical framework for integrating RPM into clinical workflows and DFU prevention strategies, drawing from their own experiences. Additionally, the study provides recommendations for clinicians on transitioning patients from remote to in-office care when needed.

Crisologo and Lavery [9], in the *Annals of Translational Medicine*, examine the potential of remote home monitoring in identifying and preventing diabetic foot ulceration. They h ighlight the crucial role of technologies like smart mats and t emperature sensors in detecting early signs of ulcers. The aut hors stress that remote monitoring can facilitate timely interv entions, reducing the incidence and severity of DFUs.

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Reyzelman et al.'s observational study [10] assesses the effectiveness of continuous temperature-monitoring socks for home use in diabetic patients. These socks, embedded with sensors, successfully monitored foot temperature and provided real-time data, aiding in early detection of issues like foot ulcers. This method offers a non-intrusive and convenient way for patients to manage their foot health at home.

Bus et al. [11] explore advancements in footwear, sensor technology, and digital health for enhancing offloading and adherence in diabetic foot care. Their narrative literature review highlights innovations such as smart insoles and remote monitoring systems that improve adherence to offloading strategies. This review emphasizes the potential of these technologies to enhance foot care outcomes for individuals with diabetes. Additionally, the management of diabetic foot ulcers with modern technology is discussed in several research papers [12-20].

The integration of hardware-based sensors with predictive analytics has proven to be a highly promising approach for early detection of foot ulcers. Researchers have highlighted the critical role of combining real-time sensor data with predictive models to create a seamless, proactive healthcare solution. This review outlines significant advancements and challenges in automated foot ulcer detection, emphasizing the transformative potential of predictive analytics and sensor technologies in preventive healthcare.

Despite recent technological advancements, challenges like data collection, processing, and optimal model selection persist. This proposed work seeks to address these research gaps, providing solutions to overcome these limitations and improve the effectiveness of automated foot ulcer detection systems.

III. METHODOLOGIES

This research paper focuses on the real-time monitoring of foot ulcer using a remote patient monitoring system. The methodology involves the integration of hardware and software technologies to achieve early diagnosis. The main objective is to prevent amputation and improve patient outcomes. A predictive analytics model integrated with hardware and software for the automatic detection of foot ulcers can revolutionize care by providing a timely and accurate diagnosis. Here's how this system can be designed:

A. Hardware Setup: The foot ulcer detection system is built using an **Arduino Uno microcontroller** connected with both a **temperature sensor** and a **pressure sensor**. These sensors are crucial for capturing relevant data about foot conditions, such as temperature variations (indicative of infection or ulcer formation) and pressure points (indicative of poor circulation or sensitivity).

B.Temperature Sensor:

- A DS18B20 temperature sensor is used to measure the surface temperature of the foot. Significant temperature differences between various regions of the foot may indicate the presence of inflammation or early-stage ulcers.
- The sensor is connected to the Arduino Uno, which processes the raw temperature data. The temperature sensor is calibrated to ensure that the readings accurately reflect the actual skin temperature.

C.Pressure Sensor:

- A FSR (Force Sensitive Resistor) pressure sensor is employed to measure the pressure exerted on different areas of the foot. Abnormal pressure distribution can indicate areas of concern that may be prone to ulceration.
- The pressure sensor is connected to the Arduino Uno, which processes the pressure data, providing insights into the foot's sensitivity and weight distribution.

D. Arduino Uno Microcontroller:

- The Arduino Uno is chosen for its versatility and ease of integration with external sensors. It is programmed using the Arduino IDE with custom code designed to manage the data collection, processing, and communication with a data analysis system.
- The Arduino reads both temperature and pressure sensor inputs, processes the data, and sends it to the software component for further analysis.

E. Calibration and Validation:

a. The setup is calibrated using clinical standards to ensure accuracy in measuring both foot temperature and pressure distribution. The sensors are validated against medical-grade devices to confirm their reliability for ulcer detection.

F. Data Communication:

- The Arduino sends the processed data to a connected computer or mobile device via a serial connection or Bluetooth module. The data is then fed into a software-based predictive analytics model for real-time ulcer detection.

G. Software Integration:

- The collected sensor data is transferred to the software, where it undergoes preprocessing and is used to build a predictive model for identifying potential ulcer formation.

By combining temperature and pressure data, the system provides a comprehensive evaluation of the foot's condition, helping to detect early signs of ulcers and prevent further complications.

H. Software Setup

The software setup for detecting foot ulcers using a pressure sensor and a temperature sensor involves several key components, including data acquisition, preprocessing, model building, and deployment for real-time predictions. Below is a step-by-step guide to the software setup:

- *Programming Environment:* The software is developed using Python for data handling, processing, and model building. Arduino IDE is used to program the Arduino Uno to collect data from the sensors.
- *Data Acquisition*: The sensor data (temperature and pressure) is collected in real-time from the Arduino and transferred to the software environment for further processing. Communication between Arduino and the

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software is established using a serial port or Bluetooth module.

- *Data Collection & Logging:* A **Python script** running on the computer will read the data from the serial port (sent by Arduino), log it, and save it in a structured format like a CSV file for further processing.
- *Data Preprocessing:* Once the data is collected, it needs to be preprocessed for model building. Common preprocessing steps include **handling missing values**, **data normalization**, and **feature extraction**.
- *Model Building (Predictive Analytics):* Use machine learning models like **Naïve Bayes, Decision tree**, **Random Forest**, and **Support Vector Machines** are to build a predictive model. The model predicts whether or not a foot ulcer is likely based on the **temperature** and **pressure** sensor data.
- *Model Validation*: Validate the model using metrics like **accuracy**, **precision**, **recall**, and **ROC-AUC score**.
- *Real-time Prediction and Alert System:* Once the model is trained and validated, it can be deployed for real-time predictions. Based on the incoming sensor data, the system can predict the likelihood of a foot ulcer and send an **alert** if the risk is high.

This research paper focuses on developing a real-time monitoring system for foot ulcers using an integrated hardware and software approach to improve early diagnosis and prevent amputations. The system combines a predictive analytics model with sensor data from a temperature sensor (DS18B20) and a pressure sensor (FSR) connected to an Arduino Uno microcontroller. This integration of hardware and software provides a comprehensive solution for early ulcer detection, enabling timely intervention and better patient outcomes.

IV. Results and Discussion

The project's methodology led to the successful develo pment and deployment of a remote patient monitoring syst em for pulse rate analysis. This section delves into a detaile d discussion of the results and their implications. The figure 4 to 10 shows the implementation results.

A. Hardware Setup

The initial stage involves setting the hardware by attaching both a pressure sensor and a temperature sensor to the Arduino Uno microcontroller. In order to accurately gather data on foot pressure and temperature—two factors that are crucial for identifying early indications of foot ulcers—this configuration was necessary. It was crucial to guarantee the dependability of these sensors and to apply calibration techniques to optimize their precision. While the temperature sensor kept an eye out for any indications of inflammation, the pressure sensor recorded sensitivity and weight distribution. This sturdy hardware configuration offered a dependable basis for further data processing and analysis.

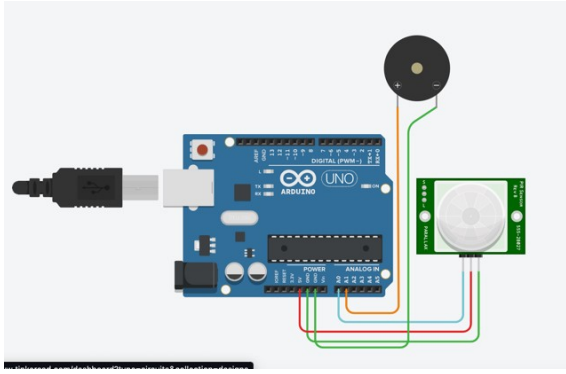

Figure 3:Sensor Integration

B. Data Processing:

Following the hardware setup, custom Arduino code was developed to process the raw data from both the pressure and temperature sensors. This code was carefully designed to ensure efficient, real-time data acquisition and processing. The collected data, which included foot pressure distribution and temperature variations, was then transmitted and converted into Python code for advanced analysis. Python's flexibility and extensive data analysis libraries made it an ideal choice for processing the sensor data. By converting the data into Python, it became compatible with various analytical tools, enabling thorough examination and interpretation of the sensor readings for early ulcer detection.

C. Dataset description

The dataset consists of Blood Sugar level, Age, Foot temperature, Ulcer size, Ulcer depth, Sensation loss (1 if there is no sensation otherwise) and Foot ulcer(target variable). This dataset is used for model building.

D. Model Building

The predictive analytics model like Decision tree classifier, Random Forest Classifier, Naïve Bayes, and comparison of metrics f1-score is shown in the following figures

```
Model Accuracy: 54.00%

Classification Report:
              precision    recall  f1-score   support

           0       0.51      0.60      0.55        94
           1       0.58      0.49      0.53       106

    accuracy                           0.54       200
   macro avg       0.54      0.54      0.54       200
weighted avg       0.55      0.54      0.54       200
```

Figure 4:Decision Tree-Classification Report

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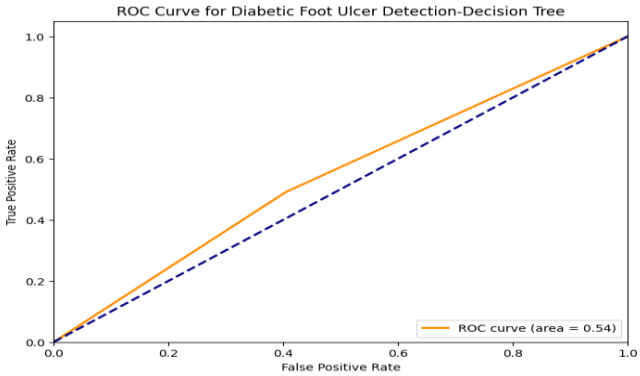

Figure 5:Decision Tree-ROC

```
Model Accuracy: 54.50%

Classification Report:
              precision    recall  f1-score   support

           0       0.52      0.49      0.50        94
           1       0.57      0.59      0.58       106

    accuracy                           0.55       200
   macro avg       0.54      0.54      0.54       200
weighted avg       0.54      0.55      0.54       200
```

Figure 6:Random Forest-Classification Report

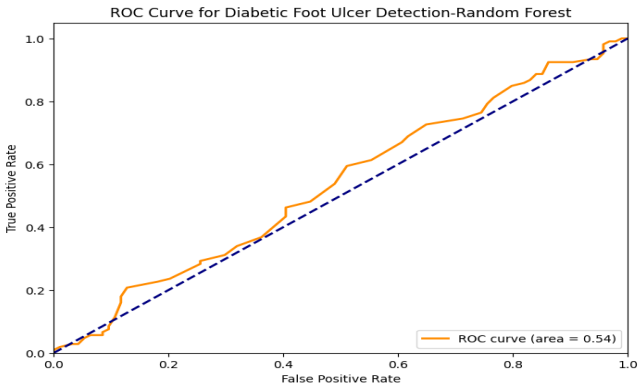

Figure 7:Random Forest-ROC

```
Classification Report:Naive Bayes
              precision    recall  f1-score   support

           0       0.51      0.30      0.38        94
           1       0.54      0.75      0.63       106

    accuracy                           0.54       200
   macro avg       0.53      0.52      0.50       200
weighted avg       0.53      0.54      0.51       200
```

Figure 8:Naive Bayes-Classification Report

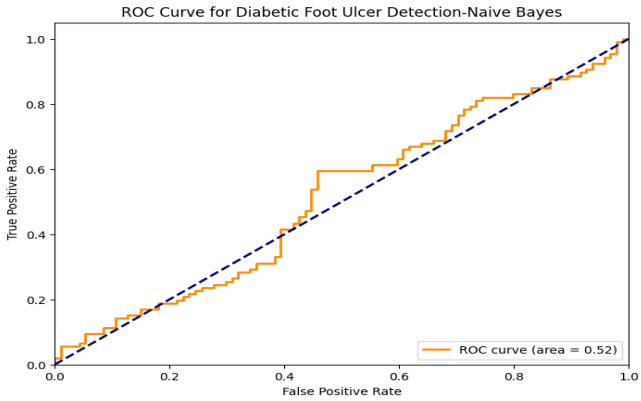

Figure 9:Naive Bayes-ROC

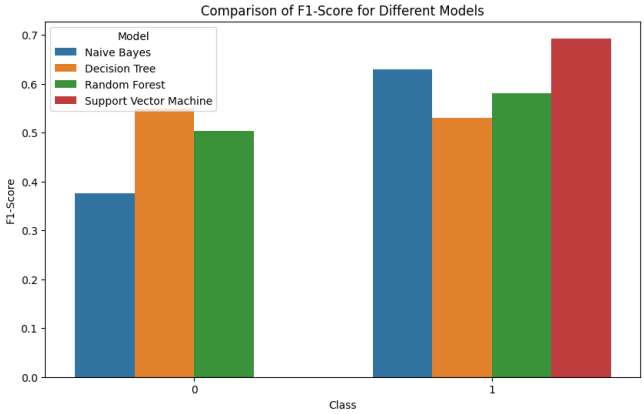

Figure 10:Comparison of F1-Score - Yes-No Class

The figure 4 to 8 shows results like accuracy, precision, recall, ROC and fi-score metrics of various predictive models.

To examine the performance of several prediction models on the dataset, like decision trees, random forests, and Naïve Bayes had been implemented. The comparison shows that, especially in terms of accuracy and resilience, the random forest method consistently performed better than the other two models. The outcomes highlight random forest's efficiency in managing intricate datasets with several variables, including blood sugar levels, foot temperature, ulcer size, and depth, among others. Because of this, random forest is a very good option for predictive analytics when it comes to identifying foot ulcers using the provided dataset.

In the real time scenario the patient details like temperature in foot is measured by temperature sensor and sensation loss is measured by the pressure sensor are collected and that information fed into the predictive model to identify the foot ulcer of a patient.

V. CONCLUSION

In this study, an automatic foot ulcer detection system that integrates hardware sensors and predictive analytics to enhance early diagnosis and improve patient outcomes. Foot ulcers, especially in diabetic patients, pose a serious health risk if not identified early. To address this, the system utilizes a foot temperature sensor and a pressure sensor to capture key indicators of ulcer formation. These sensors provide real-time data, which is then processed and analyzed by machine learning models, including Support Vector Machine, Naïve Bayes, Decision Tree, and Random Forest. Upon comparing

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the models, Random Forest consistently outperformed the others, demonstrating its effectiveness for this application. The results highlight the potential of this integrated approach to provide a non-invasive, real-time solution for early ulcer detection, which can be deployed in both clinical and home settings for high-risk patients. By enabling timely interventions, this technology promises to reduce complications, hospitalizations, and healthcare costs, ultimately improving the quality of life for patients at risk of developing foot ulcers.

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